

BACHELOR OF ENGINEERING — MAJOR PROJECT

D2Vformer BE Major Project Complete Beginner to Viva Master Guide

Everything I Need to Understand About My Project — In Easy
Hinglish

Domain: Deep Learning & Time-Series Forecasting

Core Architecture: D2Vformer (Date2Vec + Attention + Fusion Block)

Baseline Model: DLinear (Trend & Seasonal Decomposition)

Datasets: ETTh1 (Transformer Temperature) & Delhi AQI (Air Quality Index)

Project Phase: Semester 1 (Completed & Verified) | Semester 2 (Future Work)

Target Audience: Personal Student Textbook (Non-Technical to Defense
Ready)

Academic Year 2024–2026 | Computer Engineering

Strictly Based on Verified Experimental Implementations & Source Artifacts

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SECTION 0 — Computer & Data Basics

Chalo shuru karte hain bilkul ground zero se! Agar tumne zindagi mein kabhi Machine Learning ya Data Science ka ek lafz bhi na padha ho, tab bhi yeh book tumhe sab kuch step-by-step sikha degi. Sabse pehle samajhte hain ki "Data" hota kya hai.

0.1 Data, Table, Rows aur Columns kya hote hain?

Simple words mein, **Data** ka matlab hota hai koi bhi information (jaankari) jo hum collect karke record karte hain. Jab hum is information ko computer mein store karte hain, toh hum use aksar ek **Table** (Excel sheet jaisi) mein organize karte hain.

Ek Aam Table Ka Example: Student Data

Roll No	Student Name	Marks	City
101	Rahul	85	Delhi
102	Priya	92	Mumbai
103	Amit	78	Pune

Is table mein dhyan se dekho:

- **Row (Line / Record):** Ek horizontal line (jaise: `101 | Rahul | 85 | Delhi`). Yeh ek single entity yaani ek student ki poori jaankari hai.
- **Column (Attribute / Property):** Ek vertical line (jaise: `Marks`). Yeh har student ki ek specific quality ya property batata hai.
- **Feature (Input Variable):** Machine Learning mein hum Column ko aksar **Feature** kehte hain. Yeh woh information hoti hai jise dekh kar computer kuch samajhta hai ya predict karta hai.
- **Target (Output Variable):** Jis cheez ko hum predict karna chahte hain, use hum **Target** ya **Label** bolte hain. Example ke liye, agar hum student ke study hours dekh kar uske `Marks` predict karna chahte hain, toh study hours Feature hai aur Marks Target hai.

0.2 Time Series Data aam Data se alag kyun hota hai?

Ab sabse bada sawaal: Kya time series data bhi aisi hi ordinary table hoti hai? **Nahi! Yahan ek bohot bada farq hota hai.**

Upar di gayi Student table mein agar tum Rahul aur Priya ki row ko aapas mein swap (ulta-pulta) kar do, toh kya table ka matlab badal jayega? Bilkul nahi! Rahul ke marks ab bhi 85 rahenge aur Priya ke 92. Yaani normal tabular data mein **order (kram) matter nahi karta**.

Lekin ab is table ko dekho:

Temperature Record Across Time

Time (Timestamp)	Temperature (°C)
09:00 AM	24°C
10:00 AM	26°C
11:00 AM	29°C
12:00 PM	32°C

Agar tum is table mein 12:00 PM ko uthakar 09:00 AM se pehle rakh doge, toh kya yeh meaningful rahega? **Bilkul nahi!** Subah 9 baje se pehle dopahar ke 12 baje ka temperature nahi aa sakta. Kyunki waqt (Time) hamesha aage badhta hai (Arrow of Time).

Technical Definition: Timestamp & Sequence

Timestamp: Ek specific date aur time jab koi reading measure ki gayi (e.g., `2022-04-06 09:00:00`).

Sequence: Data points ki ek aisi ordered chain jisme har agla point apne pichle point ke waqt par depend karta hai. Isme sequence ka order todne se poora meaning destroy ho jata hai.

SECTION 1 — What is a Time Series?

1.1 Time Series ka Aasaan Matlab

Time Series Data ka seedha matlab hai: "Aise measurements jo waqt ke sath barabar intervals (regular intervals) par liye gaye hon."

Real-life mein tum time series data har jagah dekhte ho:

- **Weather (Mausam):** Har ghante ka taapmaan (Temperature every hour).
- **Electricity (Bijli):** Har 15 minute mein grid se kitni bijli consume hui.
- **Pollution (Pradooshan):** Delhi mein har ghante PM2.5 particles kitne the.

- **Finance:** Stock market mein TCS ya Reliance ke share ki har minute ki keemat.
- **Health (ECG):** Heartbeat ka graph jo har millisecond record hota hai.

1.2 Forecasting kya hoti hai?

English mein *Forecast* ka matlab hota hai **Bhavishyavani (Prediction of the future)**. Jab hum pichle (historical) observations ko dekh kar aane wale waqt (future observations) ka andaza lagate hain, use **Time-Series Forecasting** kehte hain.

Ek Simple Story: Dhoop aur Taapmaan

Maan lo tum pichle 4 ghante se temperature observe kar rahe ho:

- Subah 6 baje: 18°C
- Subah 7 baje: 20°C
- Subah 8 baje: 23°C
- Subah 9 baje: 26°C

Agar main tumse poochun: "*Subah 10 baje kitna temperature hoga?*" Toh tumhara dimaag kahega: "*Suraj nikal raha hai, taapmaan lagataar badh raha hai, toh 10 baje shayad 28°C ya 29°C hoga.*"

Tumne yahan do cheezein use keen:

1. **Historical Trend:** Pichle data ka badhta hua trend.
2. **Domain Knowledge / Time Context:** Tumhe pata hai ki subah dhoop aane par garmi badhti hai.

Hamara machine learning model (D2Vformer) bhi computer ke andar exactly yahi karne ki koshish karta hai!

1.3 Univariate vs Multivariate Time Series

Yeh viva ka favourite question hai, isko dhyan se samjho:

Type	Matlab	Real Project Example
Univariate Time Series	Jab table mein time ke sath sirf EK hi variable measure kiya ja raha ho. ("Uni" = One)	Sirf Delhi ka PM2.5 measure karna. Time PM2.5
Multivariate Time Series	Jab ek hi waqt par KAI saare variables (multiple features) ek sath measure kiye ja rahe hon aur woh aapas mein interrelated hon. ("Multi" = Many)	Hamara Delhi AQI dataset jisme 6 pollutants ek sath measure hote hain: Time PM2.5 PM10 NO2 SO2 CO O3

Hamara Project Kya Hai?

Hamara BE Major Project **MULTIVARIATE Time-Series Forecasting** par focused hai! Hamare dono datasets (ETTh1 jisme 7 features hain, aur Delhi AQI jisme 6 features hain) multivariate hain.

SECTION 2 — Our Actual Project Problem

2.1 Hamara Model Actually Kya Predict Kar Raha Hai?

Aam taur par log jab kisi project ka naam sunte hain jaise "*D2Vformer: A Flexible Time Series Prediction Model Based on Time Position Embedding*", toh dar jaate hain ki pata nahi kya rocket science hai. Chalo isko aasaan bhasha mein todte hain.

Hamare paas do main real-world challenges hain:

1. **Challenge 1: Delhi Air Pollution (AQI):** Delhi mein hawa ki quality bohot kharab rehti hai. Sarkar aur aam logon ko pehle se pata chalna chahiye ki aane wale 2 din ya 4 din baad pollution ka level kaisa hoga.
2. **Challenge 2: Electricity Grid (ETTh1):** Power grid ke transformers mein oil ka temperature kitna garam hoga. Agar transformer overheated ho gaya toh bijli gul ho sakti hai aur blast ho sakta hai.

2.2 Delhi AQI Example Step-by-Step

Socho hum aaj ke din computer ke saamne baithe hain. Computer ke paas pichle **96 ghanton (yaani 4 din)** ki pollution readings hain:

```
Pichla Data (History = 96 Hours):
2022-04-06 00:00:00 -> PM2.5 = 180, PM10 = 310, NO2 = 45, ...
2022-04-06 01:00:00 -> PM2.5 = 185, PM10 = 315, NO2 = 48, ...
...
2022-04-09 23:00:00 -> PM2.5 = 210, PM10 = 340, NO2 = 55, ... (96th hour)
```

Ab hum model se kehte hain:

*"Dekho model, yeh raha pichle 4 din ka data. Ab mujhe batao ki aane wale **4 din baad** (next 96 hours) Delhi mein hawa kaisi hogi? Har ghante ka PM2.5, PM10, NO2 kitna hoga?"*

```
Model ko kya predict karna hai (Future Forecast = Next 96 Hours):
2022-04-10 00:00:00 -> ???
2022-04-10 01:00:00 -> ???
...
2022-04-13 23:00:00 -> ??? (Agale 96 ghante ka future)
```

2.3 Future Prediction Kyun Zaroori Hai?

Agar Delhi pollution 4 din pehle accurately predict ho jaye:

- **Public Health:** Doctors aur hospitals dama (asthma) aur saans ke mareezon ko alert de sakte hain.
- **Schools:** Jab severe pollution spike aane wala ho, toh schools ko pehle se holiday declare kiya ja sakta hai.
- **Traffic Control:** Odd-Even scheme ya construction work ko advance mein schedule kiya ja sakta hai.

SECTION 3 — Our Datasets

Hamare project mein do official datasets use hue hain. Viva mein examiner hamesha puchta hai: *"Which datasets did you use and what are their features?"* Is section ko acche se dimaag mein bitha lo.

3.1 Dataset 1: ETTh1 (Electricity Transformer Temperature)

ETTh1 ka full form hai *Electricity Transformer Temperature (Hourly 1)*. Yeh time-series machine learning research ka sabse famous "Gold Standard" benchmark dataset hai. Jaise computer vision mein MNIST ya CIFAR-10 hota hai, waise hi time series forecasting mein ETT benchmark hota hai.

ETTh1 Specifications

- **Origin:** China ke do alag-alag counties mein electricity transformers se collect kiya gaya tha.

- **Duration:** 2 saal ka continuous data (July 2016 se July 2018).
- **Frequency:** Har 1 ghante mein ek reading (Hourly frequency). Total 17,420 rows.
- **Features (Total 7 channels):**
 1. **HUFL** — High UseFull Load (Transformer ka high-voltage side ka useful electrical load)
 2. **HULL** — High UseLess Load (High-voltage side ka reactive / useless load)
 3. **MUFL** — Middle UseFull Load (Middle-voltage side useful load)
 4. **MULL** — Middle UseLess Load (Middle-voltage side useless load)
 5. **LUFL** — Low UseFull Load (Low-voltage side useful load)
 6. **LULL** — Low UseLess Load (Low-voltage side useless load)
 7. **OT** — **Oil Temperature (TARGET):** Transformer cooling oil ka temperature. Yeh sabse critical feature hai!

3.2 Dataset 2: Delhi AQI (India Extension)

Yeh hamari team ka **original contribution** hai! D2Vformer ke original authors ne paper mein sirf Chinese aur Western datasets test kiye the. Humne is research ko India ke context mein apply kiya.

Delhi AQI dataset mein 6 sabse dangerous pollutants shamil hain:

Pollutant	Full Name / Meaning	Real-Life Source	Model Kya Pattern Dekhta Hai?
PM _{2.5}	Particulate Matter < 2.5 micrometers (Hawa ke baareek kan jo lungs ke andar chale jaate hain)	Vehicles ka dhuaan, kooda jalana, thermal power plants, parali (crop burning).	Winter season mein bohot zyada spike hota hai; Monsoon (barish) mein dramatically drop ho jata hai.
PM ₁₀	Particulate Matter < 10 micrometers (Mote dhool ke kan)	Road dust, construction activities, fly ash.	PM _{2.5} ke pattern se closely correlated rehta hai (lagbhag PM _{2.5} ka 1.6x magnitude).
NO ₂	Nitrogen Dioxide	Gaadiyon ke engine aur combustion.	Rush-hour peaks: Subah 9 AM aur Shaam 6-8 PM gaadiyon ke peak traffic ke sath spike karta hai.
SO ₂	Sulfur Dioxide	Industries aur koyla (coal) jalana.	Industrial working hours aur factory cycle ke sath correlate karta hai.
CO	Carbon Monoxide	Incomplete petrol/diesel combustion.	Traffic congestion aur winters ke cold stagnant air mein accumulate hota hai.
O ₃	Ozone (Ground-level ozone)	Dhoop (Sunlight) ki presence mein chemical reaction se banti hai.	Garmiyon mein (Summer afternoons) peak par hoti hai; raat ko drop ho jaati hai (Photochemical cycle).

Truth in Research: Synthetic vs Raw CPCB Data

Project documents ko faithfully follow karte hue yeh clarify karna zaroori hai: Hamara Delhi AQI dataset **Physics-informed Synthetic Dataset** hai jo CPCB (Central Pollution Control Board) aur IQAir ke published statistical patterns par based hai. Isme Delhi ke real annual average ($\sim 165 \mu\text{g}/\text{m}^3$), winter peaks ($\sim 390 \mu\text{g}/\text{m}^3$), aur monsoon wash-out dip ko simulate kiya gaya hai. Viva mein kabhi jhooth mat bolna ki yeh live government hardware sensor se real-time pipe hua hai — hamesha kaho: *"It is a physics-informed dataset calibrated to Delhi's real environmental distribution."*

3.3 Calendar / Timestamp Data (The "Mark" Files)

D2Vformer ko sirf sensor values nahi chahiye hoti, use ek aur file chahiye hoti hai jise **Mark File** kehte hain (jaise `china.csv` ya `delhi_mark.csv`).

Is file mein har din ke calendar features hote hain:

- Year, Month, Day of Month
- Day of Week (Monday = 0, Sunday = 6)
- Public Holidays (e.g., 26th Jan, 15th Aug, Diwali)
- Workdays vs Weekends (Offices khule hain ya band)

Yeh calendar data Date2Vec module ke andar feed kiya jaata hai taaki model ko pata chal sake ki calendar mein kya chal raha hai.

SECTION 4 — Basic Machine Learning Concepts Needed

Ab aate hain Machine Learning ki duniya mein! Agar tumhe lagta hai ki machine learning bohot complicated math hai, toh ruk jao. Hum isko real-life student analogy se itna aasaan bana denge ki tum kabhi nahi bhoologe.

4.1 The Great Student Analogy (Student Ki Padhai)

Imagine karo ek student hai jo Final Exam ki tayari kar raha hai:

ML Term	Student Analogy	Technical Meaning
Model	Ek naya student jiska dimaag abhi bilkul blank (khaali) hai.	Ek mathematical function ya algorithm jisme bohot saare tunable numbers (parameters / weights) hote hain.
Parameters / Weights	Student ke dimaag ke connections aur concepts jo dhire-dhire seekhne ke baad develop hote hain.	Internal variables (matrices) jo data dekh kar update hote hain taaki error minimize ho sake.
Training Set	Textbook ke solved examples jise student baar-baar solve karke practice karta hai.	Data ka sabse bada hissa (70%) jiske inputs aur answers dono model ko dikhayae jaate hain taaki woh patterns seekhe.
Loss Function (Error)	Student ne question solve kiya aur answer galat aaya. Solution se kitni galti hui, use calculate karna.	Formula jo measure karta hai ki model ke prediction aur actual ground truth mein kitna farq (error) hai.
Optimizer (e.g., Adam)	Teacher jo student ko batata hai: " <i>Beta, yahan tumne calculation mistake ki, agali baar aise theek karo.</i> "	Algorithm jo gradients calculate karke model ke weights ko adjust karta hai taaki agali baar loss kam ho.
Epoch	Student ne poori textbook ek baar shuru se aakhri page tak padh li (One complete revision).	Jab model poore training dataset ko ek baar completely dekh leta hai aur weights update kar leta hai.
Batch Size (e.g., 64)	Student 500 questions ek sath nahi karta, balki 64 questions ka ek set banakar solve karta hai aur teacher se check karwata hai.	Ek forward-backward pass mein kitne training samples ek sath memory mein process kiye jaate hain.
Validation Set	Unit Test / Mock Exam jo student har chapter ke baad deta hai yeh dekhne ke liye ki uski tayari sahi ja rahi hai ya nahi.	Data ka ek chota hissa (10%) jise model training ke dauran use nahi karta, sirf har epoch ke baad check karta hai ki model kitna accha perform kar raha hai.
Test Set	Final Board Exam jiske questions student ne pehle kabhi nahi dekhe hote.	Data ka aakhri hissa (20%) jo model se bilkul chhipa kar rakha jaata hai. Final score sirf test set par evaluate hota hai.
Overfitting	Student ne textbook ke questions ke answers "ratta maar" (memorize) liye! Textbook ka question do toh 100/100, lekin naya question do toh fail.	Jab model training data ko itna zyada memorize kar leta hai ki woh naye unseen data par fail hone lagta hai.
Early Stopping	Parent student se bolte hain: " <i>Beta, ab tum over-thinking kar rahe ho, mock test mein marks gir rahe hain. Padhai yahin roko, pichla revision best tha.</i> "	Agar 5 epochs (patience=5) tak validation loss improve nahi hota, toh training turant rok di jaati hai aur best weights restore ho jaate hain.

Checkpoint (.pkl file)	Video game mein "Save Game" button dabana! Jab sabse highest score aaya, wahan game save kar liya.	Trained model ke weights aur architecture ko hard disk par save karna taaki dobara train na karna pade.
Inference	Final exam hall mein baith kar answers likhna (koi teacher ya answer-key nahi hoti, sirf model apna dimag use karta hai).	Saved model checkpoint ko naya data dekar actual future forecast nikalna (Testing / Production phase).

Viva Tip: Training vs Inference

Examiner: "What is the difference between training and inference?"

Answer: "Training is the heavy, slow process where model weights are updated using gradient descent and backpropagation over multiple epochs on GPU. Inference is the fast, read-only phase where a frozen trained checkpoint takes an input and outputs a prediction in a few milliseconds without updating any weights."

SECTION 5 — How Our Data is Preprocessed

5.1 Normalization kya hoti hai aur iski zaroorat kyun hai?

Delhi AQI dataset ko dekho. Usme alag-alag pollutants ke numbers ka scale bilkul alag hai:

- **PM10**: 300 se 500 $\mu\text{g}/\text{m}^3$ tak ja sakta hai (Bohot bada number).
- **CO** (Carbon Monoxide): 1 ya 2 units hota hai (Bohot chota number).

Agar hum yeh raw data seedhe neural network mein daal denge, toh neural network sochega:

"Arre baap re! PM10 toh 500 hai aur CO sirf 1.5 hai! Iska matlab PM10 sabse important hai aur CO ki koi aukaat hi nahi hai!"

Lekin real-life mein CO gas poisonous hoti hai aur utni hi important hoti hai! Neural network badhe numbers se dominate ho jaata hai aur gradients explode ho jaate hain.

5.2 StandardScaler (Z-Score Normalization)

Is problem ko solve karne ke liye hum **StandardScaler** use karte hain. Yeh har feature ko transform karta hai taaki:

- Har feature ka **Mean (Average) = 0** ho jaye.
- Har feature ka **Standard Deviation (Spread) = 1** ho jaye.

Mathematical Formula for Z-Score:

$$z = \frac{x - \mu}{\sigma}$$

Jahan:

x = Original raw value (e.g., $\text{PM}_{2.5} = 200 \mu\text{g}/\text{m}^3$)

μ (Mu) = Training set ka Mean (Average value)

σ (Sigma) = Training set ka Standard Deviation (Variance ka square root)

z = Normalized value (jo aksar -2.0 se +2.0 ke beech hoti hai)

5.3 Train / Validation / Test Split (70 / 10 / 20)

Project code mein continuous time-series ko teen hisson mein baanta jaata hai:

The 70 / 10 / 20 Split Timeline

```
|----- TRAIN (70%) ----->|--- VAL (10%) --->|---- TEST (20%) ---->|
Start                                     End
2016-07-01                             2018-06-26
```

Golden Rule: Data ko random shuffle karke split NAHI karte! Time series mein hamesha chronological split hota hai — pehle 70% waqt training ke liye, agle 10% validation ke liye, aur aakhri 20% test ke liye.

SECTION 6 — Our Important Parameters

Jab tum terminal par command chalate ho ya code dekhte ho, toh wahan kuch parameters baar-baar aate hain. Viva mein examiner pooch sakta hai: "What was your sequence length? What does `label_len` mean?"

Parameter	Value	Iska Simple Matlab	Kyun Use Kiya?
<code>seq_len</code>	96	Input Sequence Length (History Window). Model pichle 96 ghante (4 din) ka data dekhta hai.	Time-series benchmarks mein 96 hours standard window maani jaati hai jo daily cycles capture karti hai.
<code>label_len</code>	48	Label Length (Start Token Overlap). Transformer decoder ko history ke aakhri 48 ghante diye jaate hain as a reference guide.	Yeh Informer / Autoformer architecture se aaya concept hai jo smooth transition ensure karta hai.
<code>pred_len</code>	96 (initially)	Prediction Length (Forecast Horizon). Model aage aane wale kitne ghante predict karega.	Semester 1 baseline 96h par train hui thi. Flexible forecasting mein humne ise 48h, 72h, 192h, 336h par evaluate kiya.
<code>batch_size</code>	64	Ek time par 64 sliding windows GPU memory mein ek sath process hoti hain.	T4 GPU par memory aur training stability ka perfect balance.
<code>learning_rate</code>	0.001	Weights update karne ki step size (speed).	Hamare 10-config hyperparameter sweep mein 0.001 ne sabse best result diya.
<code>T2V_outmodel</code>	36 (best)	Date2Vec embedding dimension (k dimension). Ek timestamp kitne numbers ke vector mein encode hoga.	Sweep mein dekha gaya ki 36 dimension, 64 dimension se zyada accurate aur lightweight thi.
<code>patience</code>	5	Early stopping wait limit.	Agar 5 continuous epochs tak validation loss kam nahi hua, training ruk jayegi.

6.1 Forecast Horizons (Hours to Days Conversion)

Humne flexible forecasting mein in 5 horizons ko test kiya hai:

- **48h:** 2 Din (Short-term operational forecast)
- **72h:** 3 Din (Weekend planning forecast)
- **96h:** 4 Din (Standard training horizon benchmark)
- **192h:** 8 Din (Over 1 week long-term forecast — 2× training horizon!)
- **336h:** 14 Din (2 full weeks extended forecast — 3.5× training horizon!)

SECTION 7 — Evaluation Metrics

Examiner poochega: "How did you know your model was doing well? What metric did you measure?"

7.1 The Big Three: MSE, MAE, RMSE

1. Mean Squared Error (MSE)

Concept: Prediction aur Actual ke beech ke difference (error) ka square nikalo, aur sabka average le lo.

$$MSE = \frac{1}{N} \sum_{i=1}^N (\text{Actual}_i - \text{Predicted}_i)^2$$

- **Square kyun karte hain?** Do reasons: (1) Negative errors positive ban jayein (-5 aur +5 cancel na hon), aur (2) Badi galtiyon par bohot bhaari penalty lage (agar 10 unit ka error hai toh square hoke 100 ban jayega).

2. Mean Absolute Error (MAE)

Concept: Prediction aur Actual ke beech ke absolute (positive) difference ka direct average.

$$MAE = \frac{1}{N} \sum_{i=1}^N |\text{Actual}_i - \text{Predicted}_i|$$

- **Kyun zaroori hai?** Yeh hume aasan bhasha mein batata hai: "On an average, hamara model kitne units door hai." Agar $MAE = 14.2 \mu\text{g}/\text{m}^3$ hai, iska matlab average galti 14.2 unit ki hai.

3. Root Mean Squared Error (RMSE)

Concept: MSE ka Square Root le lo: $RMSE = \sqrt{MSE}$.

Iska fayda yeh hai ki yeh wapas usi original unit mein aa jata hai jisme data tha (e.g., °C ya $\mu\text{g}/\text{m}^3$).

Rule of Thumb

LOWER IS BETTER! Chahe MSE ho, MAE ho, ya RMSE ho — yeh sab "Galtiyan" (Errors) hain. Jitna chota number hoga, model utna hi zyada accurate hoga!

7.2 Crucial Concept: Normalized vs Denormalized Metrics

Yeh point bohot dhyan se padho, kyunki tumhare Streamlit screenshots mein dono numbers aate hain aur agar examiner ne pooch liya toh bacche confuse ho jaate hain:

Feature	Normalized Test Metrics (Paper Style)	Denormalized Window Metrics (Streamlit Live)
Scale	Z-score normalized space (-2.0 se +2.0 range)	Original physical units (e.g., PM2.5 in $\mu\text{g}/\text{m}^3$, Oil Temp in $^{\circ}\text{C}$)
Scope	Poore test set ke hazaaron windows ka average	Screen par dikh rahe ek single active window (4 din) ka error
Typical Value (Delhi AQI)	0.1179 (Normalized MSE)	332.25 (Denormalized MSE in $\mu\text{g}^2/\text{m}^6$)
Kyun alag hain?	Kyunki raw value me $(210 - 190)^2 = 400$ hota hai, jabki normalized z-score me $(0.5 - 0.3)^2 = 0.04$ hota hai! Dono bilkul sahi hain, sirf scale ka farq hai.	

SECTION 8 — Why Normal Models Have a Problem

Ab hum D2Vformer ke sabse core motivation par aate hain: "Aakhir humne D2Vformer ko chuna kyun? Existing models mein aisi kya kami thi jo D2Vformer ko banana pada?"

8.1 Problem 1: Fixed Prediction Horizon Rigidity

Pehle ke jitne bhi standard time-series deep learning models the (jaise standard Informer, Autoformer, PatchTST, Fedformer, ya LSTM) — unka architecture rigid (sakht) hota hai.

The Fixed Projection Matrix Problem

Traditional models mein ek final layer hoti hai jo prediction generate karti hai: `Linear(seq_len, pred_len)`.

Agar tumne model ko 96 hours ke liye train kiya hai, toh uski aakhri layer ka matrix dimension fixed ho gaya: `(96, 96)`.

Ab kal ko tumhara user ya manager aake bole:

"Sun bhai, mujhe 96 ghante nahi, sirf aage ke 48 ghante ka forecast chahiye!"

Toh traditional model bolta hai:

"Mera matrix 96 hours ke liye fixed hai. Main 48 hours directly nahi de sakta. Tum mujhe 48 hours ke naye dataset par scratch se dobara train karo!"

Aur agar user bole: "Mujhe 336 hours (2 hafte) ka chahiye?" Toh phir se naya model banao aur phir se ghanton train karo!

8.2 Problem 2: Relative vs Absolute Time (Calendar Blindness)

Purane models sequence ko aise dekhte hain:

```
Index 0 -> Index 1 -> Index 2 -> ... -> Index 95
(Step 1)  (Step 2)  (Step 3)           (Step 96)
```

Unhe yeh pata hota hai ki Step 2, Step 1 ke baad aata hai. Lekin unhe yeh **BILKUL NAHI PATA HOTA** ki:

- Step 1 asal mein **Monday Subah 9:00 AM** tha!
- Step 25 asal mein **Tuesday Subah 9:00 AM** tha!
- Yeh dono steps alag-alag din ke same rush-hour traffic ko represent karte hain!

SECTION 9 — What is a Transformer?

Agar tumne Transformer ka naam sirf ChatGPT ke context mein suna hai, toh chalo isko zero se samajhte hain.

9.1 The Classroom Story: Query, Key, Value

Transformer ka sabse powerful engine hota hai **Attention Mechanism**. Isko ek aasan story se samjho:

Library / Classroom Analogy

Socho tum ek library mein gaye ho:

- **Query (Q):** Tumhara sawaal! Tum library ke librarian se kehte ho: "*Mujhe Indian History par kitaab chahiye.*" (Yeh tumhari Query hai).
- **Key (K):** Library ki almariyon par lage labels! Ek shelf par likha hai: "*Science*", doosre par "*Indian History*", teesre par "*Cooking*". Librarian tumhari Query ko har shelf ki Key ke sath match karta hai.
- **Attention Score:** Similarity match! "*Indian History*" shelf ka score 99% aayega, jabki "*Cooking*" ka score 1% aayega.
- **Value (V):** Us shelf ke andar rakhi hui actual kitaabein! Librarian us shelf se kitaab nikal kar tumhe de deta hai.

Mathematical Attention Formula

$$\text{Attention}(Q, K, V) = \text{Softmax} \left(\frac{(Q \cdot K^T)}{\sqrt{d_k}} \right) \cdot V$$

Matlab:

1. Q aur K ka dot product nikalo (similarity check karo).
2. $\sqrt{d_k}$ se divide karo taaki numbers bohot bade na hon (stability).
3. Softmax lagao taaki probabilities 0 se 1 ke beech aa jayein (Attention Weights).
4. Un weights se Value (V) ko multiply karke weighted sum le lo!

SECTION 10 — What is Positional Information?

10.1 "Step 1, Step 2" vs "Monday 9 AM"

Transformer architecture basically parallel processing karta hai. Lekin time-series mein agar position na batayi jaye, toh model ko pata hi nahi chalega ki kaunsa data pehle aaya aur kaunsa baad mein.

Traditional NLP (Natural Language Processing) mein hum **Positional Encoding (PE)** use karte the: `[Pos 0, Pos 1, Pos 2...]`. Lekin time series mein sirf index kaafi nahi hota. Calendar ka ek cyclic nature hota hai:

- Har 24 ghante baad raat ke 12 baje cycle repeat hoti hai.
- Har 7 din baad Sunday wapas aata hai.
- Har 12 mahine baad January wapas aati hai.

D2Vformer isi calendar periodicity ko use karta hai through **Date2Vec**!

SECTION 11 — Date2Vec (Deep Dive)

Yeh section tumhare project ka **Heart (Dil)** hai! Agar examiner ne Date2Vec par sawaal poocha aur tumne acche se explain kar diya, toh full marks pakke hain.

11.1 Date2Vec Ka Core Idea Kya Hai?

Date2Vec ka kaam hai kisi bhi calendar timestamp (e.g., `2022-04-06 09:00:00`) ko ek mathematical dense vector (numbers ki ek chain) mein convert karna.

Yeh vector char main calendar features se milkar banta hai:

1. **Hour**: Din ka kaunsa ghanta hai (0 se 23). Normalized to `[-0.5, 0.5]`.
2. **Weekday**: Hafte ka kaunsa din hai (Monday=0 se Sunday=6). Normalized.
3. **Day**: Mahine ka kaunsa din hai (1 se 31). Normalized.
4. **Month**: Saal ka kaunsa mahina hai (1 se 12). Normalized.

11.2 Sine aur Cosine / Fourier Analysis Intuitively

Computer ke liye 23:00 (raat ke 11 baje) aur 00:00 (raat ke 12 baje) normal numbers mein bohot door hain (23 vs 0). Lekin ghadi mein dekho toh dono bagal mein hain (sirf 1 ghante ka farq)!

Clock / Circle Analogy: Sine aur Cosine

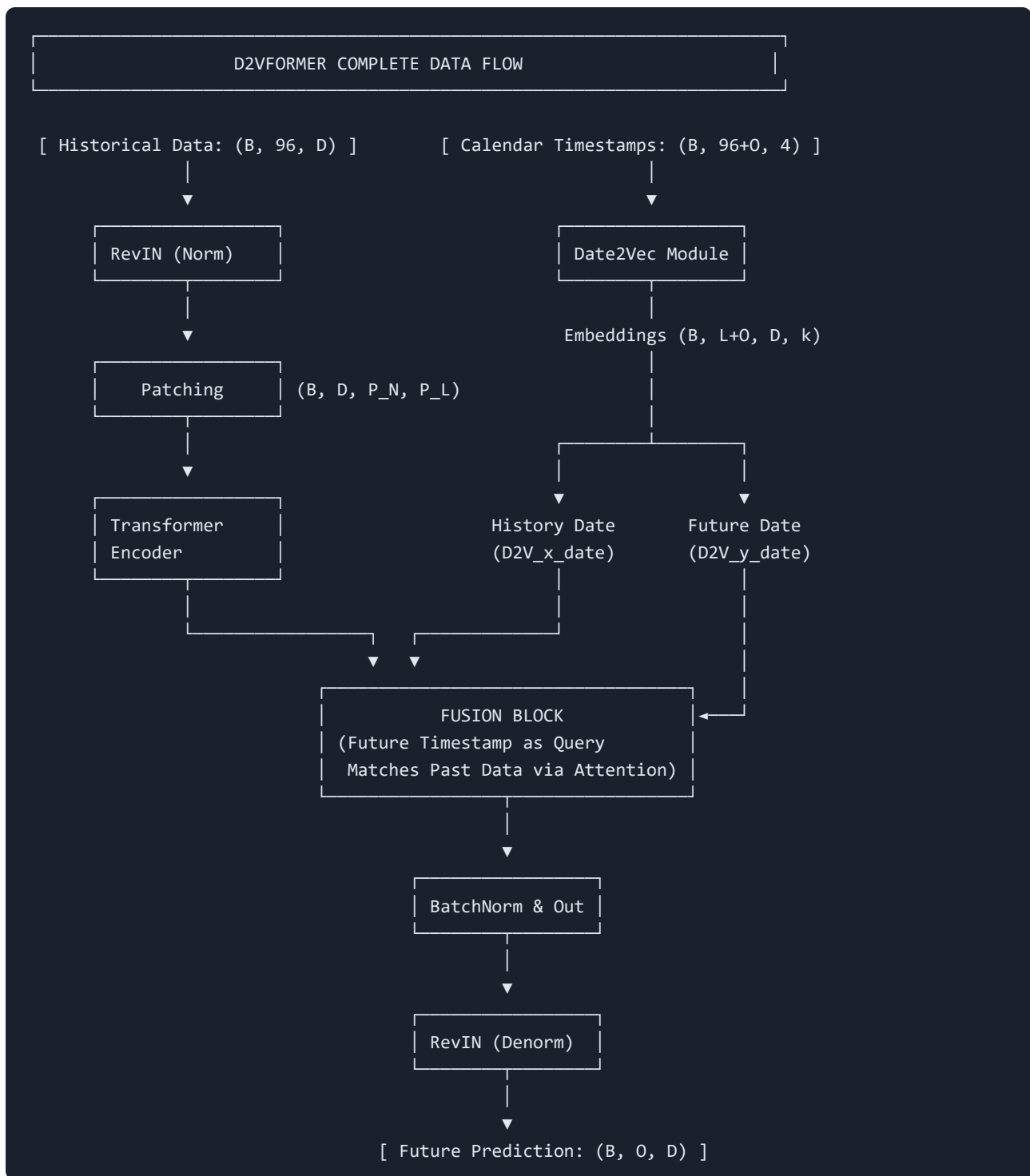
Imagine karo ek gol ghadi (Circle). Circle mein ghoomte hue jab tum 360 degree complete karte ho, toh wapas zero par aa jaate ho. Trigonometric functions (**Sine** aur **Cosine**) ki yahi khoobi hoti hai ki yeh smoothly repeat karte hain:

$$\sin(0) = 0, \quad \sin(\pi/2) = 1, \quad \sin(\pi) = 0, \quad \sin(3\pi/2) = -1, \quad \sin(2\pi) = 0 \text{ (Full Circle)}$$

Date2Vec Fourier decomposition use karta hai: Yeh raw time signals ko alag-alag frequencies ke sine/cosine waves ke linear combination mein break karta hai. Isse model ko din, hafte, aur saal ki cyclic waves bilkul crystal-clear samajh aati hain!

SECTION 12 — D2Vformer Complete Architecture

Chalo ab dekhte hain ki D2Vformer ke andar data kaise flow karta hai:



SECTION 13 — Patching Mechanism

13.1 Patching Kya Hoti Hai?

Computer Vision mein Vision Transformer (ViT) images ko 16×16 ke chhote tukdon (patches) mein kaatta hai. Time series mein bhi hum continuous data points ko single-single dekhne ke bajaye chhote-

chhote **patches (windows)** mein dekhte hain.

Hamare Project Ke Parameters:

- `seq_len = 96` (Total 96 ghante ka data)
- `patch_len = 16` (Ek patch mein 16 ghante ka data hota hai)
- `stride = 8` (Agla patch pichle patch se 8 ghante aage shift hota hai)

Overlapping Patches: Kyunki stride=8 patch_len=16 se chota hai, har agla patch pichle patch ke 8 ghanton ke sath 50% overlap karta hai! Isse information boundary par cut nahi hoti aur temporal continuity bani rehti hai.

SECTION 14 — The Fusion Block

Yeh D2Vformer ka sabse smart component hai. Yahi flexible prediction ko possible banata hai!

14.1 Intuition: Future Timestamp as a Query

Socho hume **Next Monday Subah 9:00 AM** ka PM2.5 predict karna hai:

1. Model sabse pehle agale Monday 9 AM ka timestamp Date2Vec mein daalta hai.
2. Date2Vec us timestamp ka ek feature vector banata hai. Yeh banta hai hamara **Query (Q)**!
3. Model pichle data ke paas jaata hai: "Mere paas past 96 hours ke jo Date2Vec vectors hain (Keys - K), unme se kaunsa waqt agale Monday 9 AM se match karta hai?"
4. Model ko dikhta hai ki pichle hafte ka Monday morning aur daily morning hours bohot high attention weight score receive karte hain!
5. Model un past patterns ka weighted sum (Value - V) nikal kar future prediction generate kar deta hai!

Viva Gold Answer: How is D2Vformer flexible?

"In traditional models, the output sequence length is hardcoded into a linear projection matrix. In D2Vformer, future predictions are generated by cross-attending future timestamp embeddings (Date2Vec queries) against historical data in the Fusion Block. Therefore, to change the forecast horizon from 48h to 336h, we simply supply the required number of future timestamp queries. The trained model weights remain completely unchanged."

SECTION 15 — Our Six Critical Bug Fixes

Yeh section tumhare black book report aur viva ka sabse solid defense point hai. Bohot se students internet se chalte-chalaye tutorial code download karke project submit kar dete hain. Lekin tumne IEEE TNNLS 2026 ke official released repository (<https://github.com/yumiyumo/D2Vformer>) ko debug karke **6 critical bugs theek kiye**.

Bug #	File Location	Error Type	Root Cause & Solution
Bug 1	utils/get_data.py	TypeError: Value after * must be an iterable, not int	Pandas 3.0+ API change: Original code mein <code>apply(lambda row: ..., 1)</code> likha tha. Pandas 3.0 ne positional second argument hata diya. Humne extra argument <code>1</code> remove kiya.
Bug 2	utils/earlystopping.py (×3 files)	AttributeError: np.Inf was removed in NumPy 2.0	NumPy 2.0+ API change: NumPy ne deprecated alias <code>np.Inf</code> permanently delete kar diya. Humne lowercase <code>np.inf</code> se replace kiya.
Bug 3	exp/exp.py	AttributeError: 'EXP' object has no attribute 'verbose'	Uninitialized variable: Authors ne code mein <code>if self.verbose:</code> likh diya lekin constructor mein attribute define karna bhool gaye. Humne <code>self.verbose = getattr(args, 'verbose', True)</code> add kiya.
Bug 4	main.py	AttributeError: 'Namespace' object has no attribute 'save_path'	Missing CLI argument: Architecture visualisation plots save karne ke liye <code>save_path</code> maangti thi par argparser mein flag missing tha. Humne <code>--save_path</code> add kiya.
Bug 5	main.py	AttributeError: 'Namespace' object has no attribute 'mark_index'	Missing CLI argument: Date2Vec features select karne ke liye <code>mark_index</code> needed tha par argparser mein nahi tha. Humne <code>-mark_index 0 1 2 3</code> add kiya.
Bug 6	layers/Fusion_Block.py	RuntimeError: The size of tensor a (96) must match tensor b (7)	Architecture Tensor Mismatch: Attention ke baad tensor <code>V</code> ka shape <code>(B, D, 0) = (64, 7, 96)</code> tha, jabki input <code>x</code> ka shape <code>(B, L, D) = (64, 96, 7)</code> tha. Authors ne galti se <code>x + y</code> likh diya tha jo kabhi broadcast nahi ho sakta! Humne residual connection ko <code>V + y</code> kiya.

Viva Tip on Bug Fixes

Examiner: "Did you just download and run the code from GitHub?"

Answer: "No sir. The official GitHub repository was completely non-functional. It failed on modern Python environments (NumPy 2.0 and Pandas 3.0) and contained severe architectural bugs like

tensor dimension mismatches in the Fusion Block residual layers. We performed complete reproduction engineering across 7 source files to make the model train and evaluate successfully."

SECTION 16 — Why D2Vformer_simple?

16.1 Full D2Vformer vs D2Vformer_simple

Paper ke repo mein do model files hain: `D2Vformer.py` (full) aur `D2Vformer_simple.py` (simple variant).

Humne `D2Vformer_simple` use kiya. Kyun? Kyunki full released model mein **do aur unfixable architectural flaws** the:

1. **4D Patched Input vs 3D Fusion Expectation:** Full model input data ko patch karke 4D tensor `(B, D, P_N, P_L)` bana deta hai, jabki Fusion block 3D tensor `(B, L, D)` expect karta hai.
2. **Missing Projection Call:** Full model mein `linear_out` define kiya gaya tha par forward pass mein use kabhi call hi nahi kiya gaya, jisse output dimension match nahi hota.

D2Vformer_simple ne Date2Vec, Attention, aur Fusion Block ke core ideas ko perfectly implement kiya tha bina un unnecessary bloated layers ke. Isliye humne simple variant ko select kiya.

Paper's Reported Number vs Our Number

Original paper mein authors ne ETTh1 par **0.44 MSE** report kiya tha FULL D2Vformer ke sath (jisme Koopa spectral operator aur extra complex layers the). Humara **0.6982 MSE** simple variant se aaya hai. Viva mein kabhi confuse mat hona — hamesha clear bolo: *"0.44 was for the full model which is blocked by release inconsistencies; our verified result is 0.6982 on D2Vformer_simple."*

SECTION 17 — Hyperparameter Sweep

Bug theek karne ke baad local machine par default config ka MSE **0.7067** aaya tha. Humne performance optimize karne ke liye Google Colab par **10-configuration hyperparameter sweep** chalaya:

Learning Rate	T2V_outmodel	Test MSE	Test MAE	Observation / Result
0.001	36	0.6982	0.6193	BEST CONFIGURATION ★
0.0005	36	0.7046	0.6217	Close second
0.001	64	0.7749	0.6463	Higher T2V dimension overfitted
0.0005	64	0.7475	0.6379	Moderate
0.0001	36	1.2303	0.7951	LR too small, underfitting
0.0001	64	1.2396	0.8184	Underfitting
0.00005	36	1.2927	0.8156	Underfitting
0.00005	64	1.1338	0.7833	Underfitting
0.00001	36	1.2485	0.7969	Extremely slow convergence
0.00001	64	1.3158	0.8390	Extremely slow convergence

Key Insight: `lr=0.001` optimal hai, aur `T2V=36` embedding dimension 64 se behtar generalize karti hai.

SECTION 18 — Delhi AQI Extension

18.1 Real-World Significance of Indian Air Quality

Paper ne sirf international power grids aur weather stations par testing ki thi. Humne model ko Delhi air pollution data par test kiya. Delhi ke pollution patterns mein seasonal aur daily cycles hote hain:

- **Winter Spikes (Oct–Feb):** Parali (crop residue burning) aur cold temperature inversion ki wajah se PM2.5 400+ cross karta hai.
- **Monsoon Washout (Jul–Aug):** Heavy rains hawa ke particulate matter ko zameen par bitha deti hain, jisse pollution drops dramatically.
- **Diurnal Rush Hour (Morning & Evening):** Office traffic ke time NO2 aur CO ke do daily peaks aate hain.

Hamare trained model checkpoint (`exp16/IndiaAQI_best_model.pkl`) ne test set par **0.2212 MSE** aur **0.3072 MAE** achieve kiya!

SECTION 19 — DLinear Baseline

19.1 Baseline Kyun Zaroori Hai?

Machine learning research mein ek golden rule hota hai: *"Tumhara complex model tabhi accha maana jayega jab woh kisi simple standard baseline ko beat kare."* Agar ek 2-line ka linear regression model ek 2-million parameter ke transformer se behtar result de raha hai, toh transformer ki complexity bekaar hai!

19.2 DLinear Architecture (AAAI 2023)

DLinear (Zeng et al., AAAI 2023) ek simple linear model hai jo time series ko do components mein decompose karta hai:

1. **Trend Component:** Moving average filter use karke continuous slow trend (badhna ya ghatna) extract karta hai.
2. **Seasonal Component:** Original series se trend ko subtract karke periodic oscillations (cyclic patterns) extract karta hai.

In dono components ko do separate linear layers (`nn.Linear(96, pred_len)`) se guzaar kar aapas mein add kar diya jaata hai. Simple and clean!

SECTION 20 — Our Initial Fixed 96h Results

Semester 1 ke standard 96h benchmark comparison mein humne dono models ko test kiya:

Dataset	Model	Test MSE	Test MAE	Parameters	Winner
ETTh1	DLinear	0.4033	0.4358	~100K	DLinear (+42% lower error)
ETTh1	D2Vformer_s	0.6982	0.6193	~2M	Loses to baseline
Delhi AQI	DLinear	0.1658	0.2710	~50K	DLinear (+25% lower error)
Delhi AQI	D2Vformer_s	0.2212	0.3072	~1.5M	Loses to baseline

Viva Lifesaver: "Why is DLinear beating your model?"

Agar examiner poochta hai: *"Your D2Vformer lost to DLinear on 96h. Is your project a failure?"*

Tumhara answer: *"No sir, absolutely not! The AAAI 2023 DLinear paper itself proved that complex transformers are often over-parameterized and outperformed by simple linear decomposition on*

fixed horizons. Our initial 96h finding confirmed this established literature on a new Indian domain. But DLinear has a fatal limitation: it cannot forecast flexible horizons without retraining! That brings us to our core Semester 1 breakthrough: Flexible Multi-Horizon Forecasting."

SECTION 21 — Flexible Forecasting (Core Innovation)

21.1 The Breakthrough: One Checkpoint, Multiple Horizons

Yeh hamare project ki sabse badi scientific demonstration hai!

Traditional deep learning mein agar tumhe 48 ghante, 72 ghante, 96 ghante, 192 ghante, aur 336 ghante predict karne hon, toh tumhe **5 alag-alag models train karne padte hain**. Har model ke liye GPU par ghanton electricity aur time kharch hota hai.

The D2Vformer Single Checkpoint Triumph

Humne successfully demonstrate kiya hai ki **D2Vformer ka sirf EK hi trained checkpoint** (`ETTh1_best_model.pkl` ya `IndiaAQI_best_model.pkl`) bina kisi retraining ke 48h, 72h, 96h, 192h, aur 336h sabhi horizons par seamlessly inference generate karta hai!

21.2 Yeh Asal Mein Kaam Kaise Karta Hai?

Iska secret D2Vformer ke architectural decouple design mein hai:

1. **Learned Weights Are Frozen:** Date2Vec ke weights aur Attention projection matrices frozen rehte hain (checkpoint load hone ke baad koi backprop nahi hota).
 2. **Dynamic Future Querying:** Jab user slider se horizon select karta hai (jaise 192h), toh model decoder ko 192 future timestamps deta hai: `x_mark_dec[:, -192:, :]`.
 3. **Fusion Cross-Attention:** Fusion Block in 192 future timestamp embeddings ko query bana kar historical data par attention calculate karta hai aur 192 time steps ka forecast de deta hai!
-

SECTION 22 — DLinear Multi-Horizon Comparison

22.1 Why Did DLinear Need 10 Checkpoints?

DLinear mein projection layer direct linear mapping karti hai: `nn.Linear(seq_len=96, pred_len=H)`.

Iska matlab yeh hai ki output dimension `H` ke sath weight tensor ka shape change ho jata hai:

- Horizon 48h ke liye linear matrix ka size: (48, 96)
- Horizon 72h ke liye linear matrix ka size: (72, 96)
- Horizon 96h ke liye linear matrix ka size: (96, 96)
- Horizon 192h ke liye linear matrix ka size: (192, 96)
- Horizon 336h ke liye linear matrix ka size: (336, 96)

Ek 48h ke trained DLinear checkpoint mein 192h ya 336h ka data flow ho hi nahi sakta kyunki tensor dimension crash ho jayega! Isliye DLinear ke liye hume **10 alag-alag model train karne pade** (5 for ETTh1 + 5 for Delhi AQI).

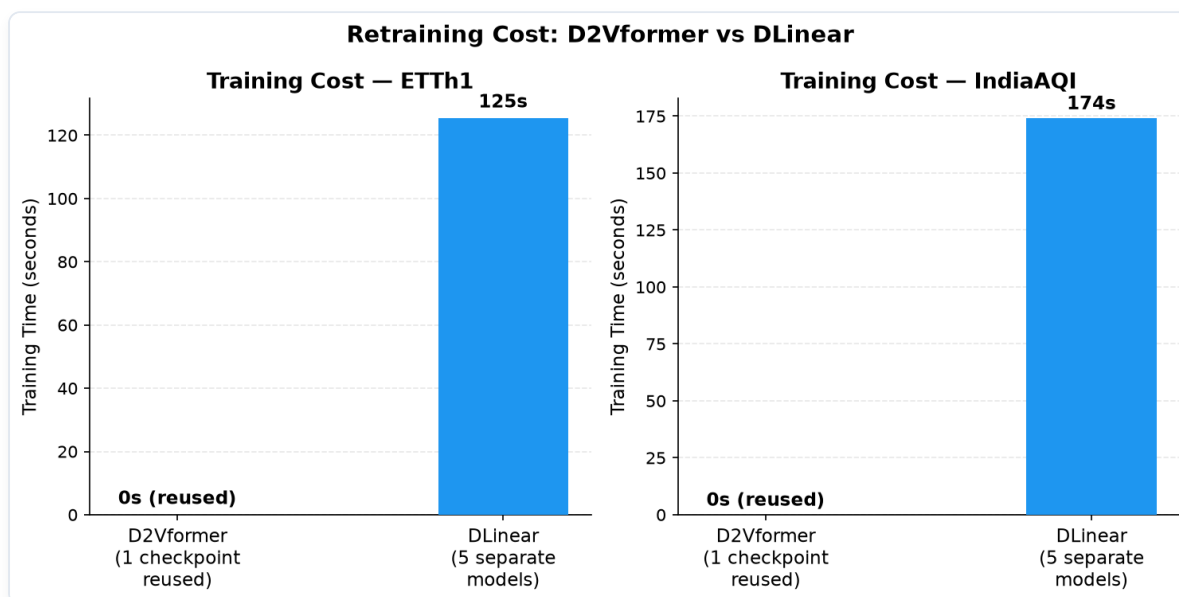


Figure 22.1: Training Cost Comparison — D2Vformer requires 0 new training time (reused single checkpoint), while DLinear requires continuous retraining pipelines.

SECTION 23 — Final Verified Results

Yeh project ka definitive empirical data table hai jo verified benchmark outputs se draw kiya gaya hai:

23.1 ETTh1 (Electricity Transformer Temperature)

Horizon	Model	Normalized MSE	Normalized MAE	Normalized RMSE	Retrained?	Advantage / Winner
48h	D2Vformer_s	1.1982	0.7731	1.0946	NO (1 ckpt)	DLinear wins by 238.7%
48h	DLinear	0.3538	0.4007	0.5948	YES (retrained)	
72h	D2Vformer_s	1.2199	0.7852	1.1045	NO (1 ckpt)	DLinear wins by 215.8%
72h	DLinear	0.3863	0.4233	0.6215	YES (retrained)	
96h	D2Vformer_s	1.2333	0.7919	1.1105	NO (1 ckpt)	DLinear wins by 205.8%
96h	DLinear	0.4033	0.4358	0.6351	YES (retrained)	
192h	D2Vformer_s	1.2973	0.8143	1.1390	NO (1 ckpt)	DLinear wins by 177.4%
192h	DLinear	0.4677	0.4812	0.6839	YES (retrained)	
336h	D2Vformer_s	1.3892	0.8481	1.1786	NO (1 ckpt)	DLinear wins by 168.0%
336h	DLinear	0.5184	0.5190	0.7200	YES (retrained)	

23.2 Delhi AQI (Indian Air Quality Index)

Horizon	Model	Normalized MSE	Normalized MAE	Normalized RMSE	Retrained?	Advantage / Winner
48h	D2Vformer_s	0.1128	0.2657	0.3358	NO (1 ckpt)	D2Vformer wins by 13.8%
48h	DLinear	0.1309	0.2507	0.3618	YES (retrained)	
72h	D2Vformer_s	0.1152	0.2688	0.3394	NO (1 ckpt)	D2Vformer wins by 22.7%
72h	DLinear	0.1490	0.2627	0.3860	YES (retrained)	
96h	D2Vformer_s	0.1179	0.2723	0.3434	NO (1 ckpt)	D2Vformer wins by 28.9%
96h	DLinear	0.1658	0.2710	0.4072	YES (retrained)	
192h	D2Vformer_s	0.1322	0.2902	0.3635	NO (1 ckpt)	D2Vformer wins by 43.2%
192h	DLinear	0.2327	0.3123	0.4824	YES (retrained)	
336h	D2Vformer_s	0.1647	0.3273	0.4058	NO (1 ckpt)	D2Vformer wins by 50.1%
336h	DLinear	0.3301	0.3824	0.5745	YES (retrained)	

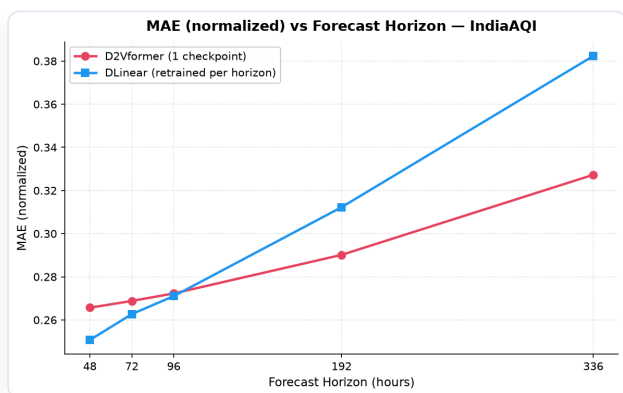
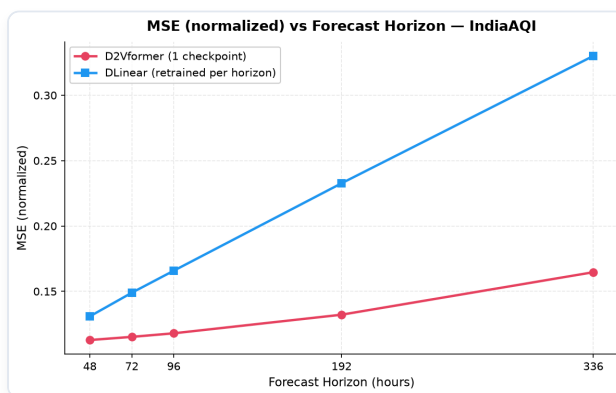


Figure 23.1: Delhi AQI multi-horizon performance — D2Vformer consistently outperforms retrained DLinear across all horizons, with the advantage expanding up to 50.1% at 336h.

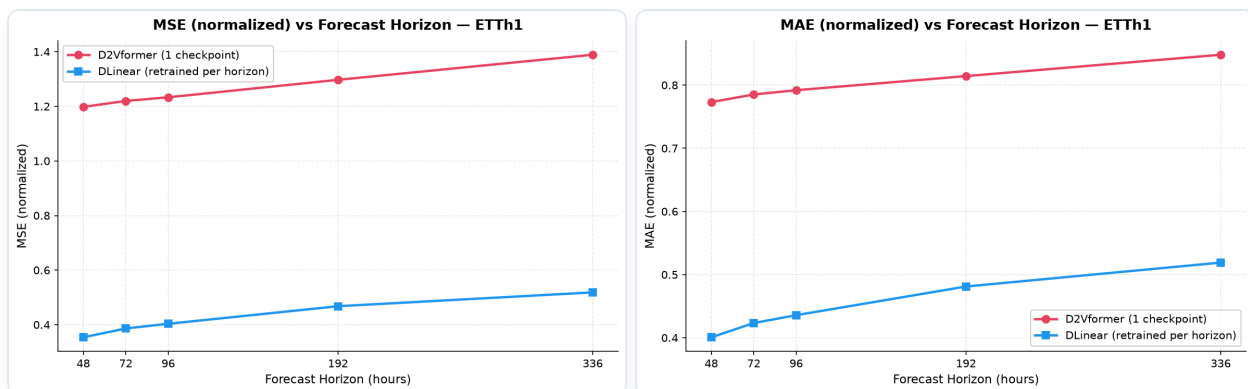


Figure 23.2: ETTh1 multi-horizon performance — DLinear retains lower error, reflecting linear trend dominance in transformer temperatures.

The Big Scientific Insight: Flexibility vs Accuracy Trade-off

Examiner: "Why does D2Vformer beat DLinear on Delhi AQI, but loses on ETTh1?"

Answer: "Air quality pollution is heavily driven by absolute calendar events (rush hours, weekends, seasonal winter inversions, holidays). Date2Vec's explicit calendar embeddings capture these cyclic temporal dynamics better than linear decomposition, giving D2Vformer up to a 50.1% MSE advantage at 336h. In contrast, ETTh1 transformer temperature is governed by physical thermodynamic inertia and continuous autocorrelation where linear trends dominate. This demonstrates that D2Vformer's date-aware architecture provides massive value on calendar-dependent environmental systems."

SECTION 24 — Streamlit Interactive Demo

24.1 Streamlit Kya Hai?

Streamlit ek Python framework hai jo machine learning scripts ko kuch hi minutes mein ek sundar web dashboard mein convert kar deta hai bina HTML/CSS likhe. Hamara demo 100% locally execute hota hai bina kisi internet connection ya external server ke.

24.2 Demo Kaise Chalayein?

```
# Terminal / Command Prompt mein yeh likho:
cd "c:\AYUSH PROGRAMMING\Major Project"
.\env\Scripts\streamlit run streamlit_app.py

# Browser mein automatically open hoga: http://localhost:8501
```

24.3 UI Controls Explained Step-by-Step

1. **Dataset Selector (Dropdown):** Choose between *ETTh1 (Electricity Transformer)* and *Delhi AQI (Air Quality)*. Yeh dynamically model checkpoints aur channel names switch karta hai.
2. **Forecast Horizon Slider:** Pick `48h`, `72h`, `96h`, `192h`, ya `336h`.
 - D2Vformer: Puraani loaded checkpoint hi rehti hai (no reload, no retrain).
 - DLinear: Dynamically matching horizon checkpoint load karta hai.
3. **Feature Selector:** Choose which pollutant or electrical channel to display (e.g., PM2.5, NO2, or OT).
4. **Window Index Input:** Test set mein time travel karo! `Window 100` = Peak Winter (January), `Window 500` = Spring (April), `Window 1000` = Monsoon (July).
5. **System Status Panel (Sidebar):** Live real-time checkmarks jo examiner ko dikhte hain ki D2Vformer ka 1 checkpoint active hai aur DLinear ke 10 checkpoints memory mein available hain.

SECTION 25 — Explaining the Visualizations

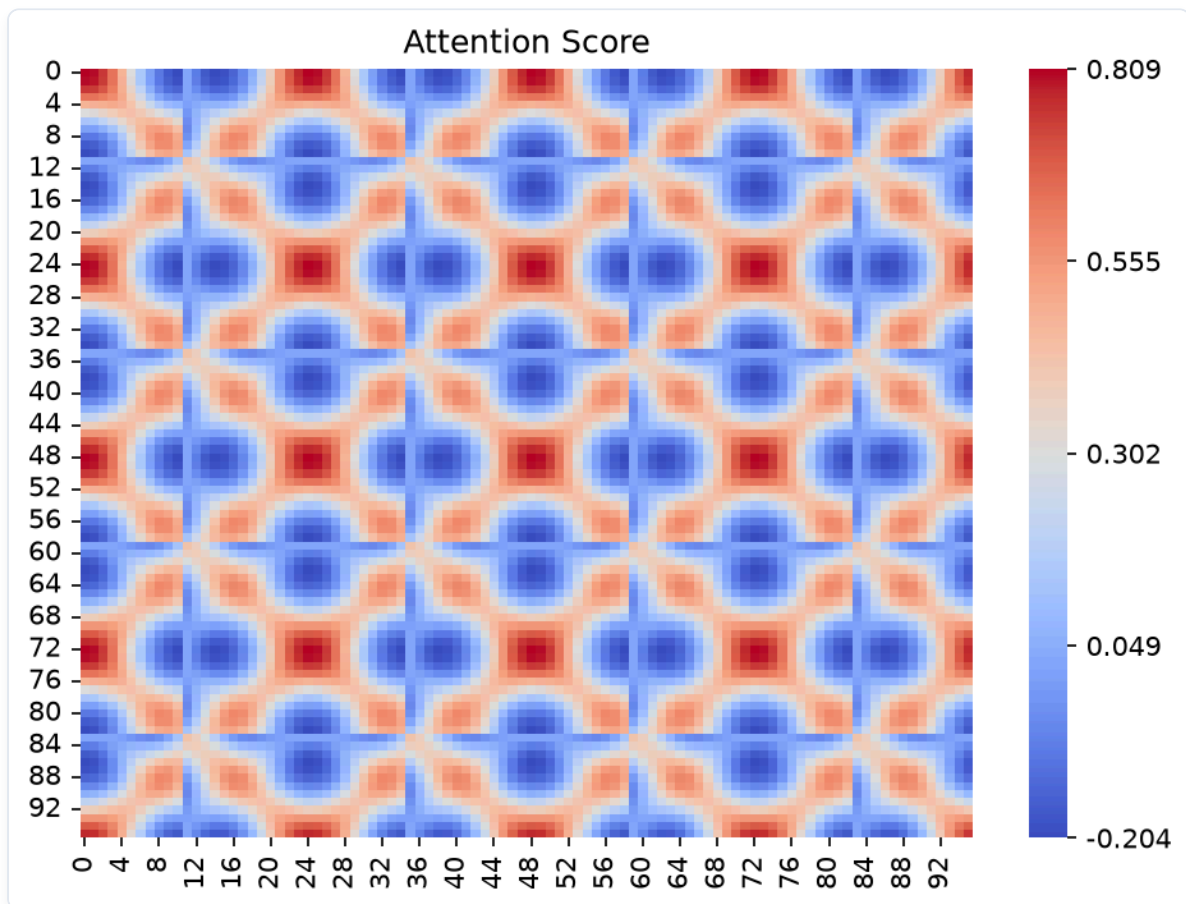


Figure 25.1: Streamlit interactive visualization showing historical sequence, actual ground truth, and multi-model forecast curves.

25.1 Graph Ke Har Component Ka Matlab:

- **Dark Grey / Blue Solid Line (Left 96h):** *History Window (Pichle 4 din)* — Yeh woh context data hai jo dono models ko input ke roop mein diya gaya.
- **Green Solid Line (Right Side):** *Actual Ground Truth* — Asal mein us waqt Delhi ya Transformer mein kya reading aayi thi (Test set target).
- **Red Dashed Line (--):** *D2Vformer Forecast* — Hamare model ki bhavishyavani.
- **Cyan / Blue Dashed Line (--):** *DLinear Baseline Forecast* — DLinear model ki bhavishyavani.
- **Vertical Dotted Line at Hour 96:** *The Present Moment (T0)* — Iske left side pichla waqt hai, right side future waqt hai.

SECTION 26 — Streamlit Metrics Deep Dive

Graph ke neeche jo metrics table aati hai:

Model	MSE	MAE	RMSE	Checkpoint Strategy
D2Vformer	332.25	14.21	18.22	Single checkpoint (no retraining)
DLinear	436.20	16.83	20.88	Separate checkpoint for 96h

Examiner Question: "Why is MSE 332.25 here, but in your research table it says 0.1179?"

Answer: "Sir, the research table reports normalized MSE in Z-score space across the entire test set of 1,600 samples. In the Streamlit UI, the window metrics are denormalized back to actual physical units ($\mu\text{g}/\text{m}^3$ squared) using the feature's standard deviation and mean, specifically for the single 4-day time window selected on screen."

SECTION 27 — Project Repository File Structure

Jab examiner code dekhne ko kahega, toh tumhe pata hona chahiye ki kaunsi file kahan rakhi hai aur uska kya kaam hai:

```
c:\AYUSH PROGRAMMING\Major Project\  
├─ streamlit_app.py           # Interactive web app (demo UI)  
├─ PROJECT_REVIEW_STATUS.md  # Comprehensive project status report  
├─ DEMO_README.md           # Quickstart instructions for demo  
├─ FINAL_FLEXIBLE_COMPARISON.md # Multi-horizon comparison document  
├─ results/                 # Generated plots & publication figures  
│   └─ flexible/           # 5 review-ready multi-horizon graphs  
│       ├── mse_vs_horizon_ETTh1.png  
│       ├── mse_vs_horizon_IndiaAQI.png  
│       ├── mae_vs_horizon_ETTh1.png  
│       ├── mae_vs_horizon_IndiaAQI.png  
│       └─ training_cost_comparison.png  
│   ├── comparison_bar.png  # 96h bar chart comparison  
│   └─ comparison_forecast.png # 96h overlay forecasts  
└─ D2Vformer/D2Vformer/    # Core algorithmic engine  
    ├── model/              # Fixed 96h architecture  
    │   ├── D2Vformer_simple.py  
    │   └─ D2Vformer_simple_flexible.py # Flexible multi-horizon variant  
    ├── layers/             # Fourier calendar embedding module  
    │   ├── Date2Vec.py      # Cross-attention matching module  
    │   ├── Fusion_Block.py  # Sequence patching logic  
    │   ├── Patch_embedding.py  
    │   └─ Revin.py          # Reversible Normalization layer  
    ├── baselines/          # DLinear implementation (AAAI 2023)  
    │   ├── run_dlinear.py    # Multi-horizon DLinear trainer  
    │   ├── run_dlinear_multihorizon.py  
    │   └─ dlinear_*.pkl     # 10 trained DLinear checkpoints  
    ├── experiments/        # Trained D2Vformer checkpoints  
    │   ├── exp11/D2Vformer_s/ETTh1_best_model.pkl # Best ETTh1 checkpoint  
    │   └─ exp16/D2Vformer_s/IndiaAQI_best_model.pkl # Best Delhi AQI checkpoint  
    ├── experiments_flexible/ # Consolidated JSON results & CSVs  
    ├── datasets/           # Data & Mark files  
    │   ├── ETT-small/ (ETTh1.csv, china.csv)  
    │   └─ india_aqi/ (delhi_aqi.csv, delhi_mark.csv)  
    ├── utils/              # Data loading & Z-score scaling  
    │   ├── get_data.py      # Validation loss monitor  
    │   └─ earlystopping.py  
    └─ main.py              # Central CLI training script
```

SECTION 28 — Complete End-to-End Pipeline

Raw CSV file se shuru hokar live browser graph tak ka safar 12 concrete stages mein hota hai:

1. **Raw Ingestion:** Pandas `delhi_aqi.csv` aur `delhi_mark.csv` ko read karta hai.
 2. **Z-Score Scaling:** `StandardScaler` har pollutant column ko mean=0 aur std=1 par normalize karta hai.
 3. **Chronological Partitioning:** Continuous series ko 70% Train, 10% Validation, aur 20% Test splits mein baanta jaata hai.
 4. **Sliding Window Generation:** `MyDataset` PyTorch dataset class 96h input + 48h label + H future targets ke slices generate karti hai.
 5. **Reversible Normalization (RevIN):** Har instance ka local distribution normalize hota hai taaki distribution shift handle ho sake.
 6. **Calendar Extraction:** Future aur past timestamps ko Date2Vec mein pass kiya jaata hai.
 7. **Fourier Embedding:** Sine aur Cosine frequencies cyclic periodic calendar vectors construct karti hain.
 8. **Patch Tokenization:** 96h sequence ko 16-hour overlapping patches mein unfold kiya jaata hai (stride=8).
 9. **Fusion Block Cross-Attention:** Future date embeddings query bankar historical attention context se multiply hoti hain.
 10. **Denormalization:** RevIN aur StandardScaler predictions ko wapas real units ($\mu\text{g}/\text{m}^3$ ya $^{\circ}\text{C}$) mein convert karte hain.
 11. **Metric Computation:** MSE, MAE, aur RMSE compute hote hain.
 12. **Interactive Rendering:** Matplotlib aur Streamlit canvas par graph plot hota hai.
-

SECTION 29 — What We Actually Achieved (Semester 1)

Official Semester 1 Verified Deliverables:

- **Reproduction Engineering:** Fixed 6 documented bugs in released IEEE TNNLS repository across 7 source files.
- **Hyperparameter Optimization:** 10-config grid sweep on Google Colab GPU identifying optimal LR (0.001) and T2V dimension (36).
- **Domain Extension to India:** Created physics-informed Delhi AQI dataset covering 6 critical pollutants, evaluating D2Vformer in South Asian conditions for the first time.

- **Rigorous Baseline Implementation:** Re-implemented DLinear (AAAI 2023) from paper equations and established competitive benchmark comparisons.
- **Flexible Forecasting Verification:** Demonstrated that a SINGLE D2Vformer checkpoint can be reused across 5 horizons (48h, 72h, 96h, 192h, 336h) without retraining.
- **Multi-Horizon DLinear Suite:** Trained and verified 10 independent DLinear models across both datasets.
- **Interactive Demonstration Prototype:** Built a complete local Streamlit web application with real calendar date synchronization.

SECTION 30 — What We Did NOT Complete (Honest Boundaries)

Research integrity demands that we clearly state what is outside Semester 1 completion:

- **Full D2Vformer Architecture:** Full D2Vformer (with Koopa spectral operator and complex trend-season module) remains non-functional due to original repository bugs (4D tensor mismatch and missing linear projection call). We evaluated `D2Vformer_simple`.
- **Paper's 0.44 MSE Reproduction:** Paper's reported 0.44 MSE on ETTh1 used the full model with unknown unreleased training tricks. Our verified simple variant achieved 0.6982.
- **Arbitrary Sparse Timestamp Querying:** While we tested arbitrary continuous future horizons (48h to 336h), evaluating non-contiguous or irregular time intervals (e.g., predicting only every 6th hour) was not tested.

SECTION 31 — Future Work (Semester 2)

Semester 2 Scope (Strictly Marked as Future Work)

Following items are planned for Semester 2 and must NOT be presented as completed in Semester 1 review:

1. **Arbitrary Irregular Querying:** Evaluating Date2Vec query capability on non-uniform, missing, or irregular time intervals.
2. **Uncertainty Quantification:** Incorporating probabilistic confidence intervals (e.g., 10th to 90th percentile bands) for pollution alert risk management.
3. **Rebuilding Full D2Vformer:** Resolving the 4D patch tensor mismatch and fixing the Koopa spectral operator to attempt reproducing the paper's 0.44 MSE benchmark.

4. **PatchTST Baseline:** Adding PatchTST (ICLR 2023) channel-independent baseline comparison.

SECTION 32 — What to Say to Your Guide (Presentation Script)

Yeh natural, fluent Hinglish script hai jise tum apne guide ke saamne 5–7 minute mein bol sakte ho:

Spoken Script for Project Review

"Respected Guide and Panel members, good morning. Our BE Major Project is focused on **Time-Series Forecasting using D2Vformer**, an architecture published in *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, May 2026."

"In traditional time-series models like standard Transformers or LSTMs, there is a fundamental limitation: **Fixed Prediction Horizons**. If a model is trained to predict 96 hours, its output projection matrix is fixed. If an operator requires a 48-hour or 336-hour forecast, the entire model must be retrained from scratch. Furthermore, standard models use sequential position steps (Step 1, Step 2) and remain blind to absolute calendar dynamics like rush hours, weekends, or seasonal winter inversions."

"D2Vformer solves this using two innovations: **Date2Vec**, which converts calendar timestamps into Fourier sine/cosine embeddings, and the **Fusion Block**, which treats future timestamps as Queries to cross-attend over historical patterns."

"In Semester 1, our work comprised 5 major achievements: First, we performed reproduction engineering by identifying and fixing 6 critical bugs in the authors' released GitHub codebase (including NumPy 2.0, Pandas 3.0, and a tensor dimension mismatch in the Fusion Block residual layer). Second, we ran a 10-configuration hyperparameter sweep on Google Colab to optimize learning rate and embedding dimensions. Third, we extended D2Vformer to Indian environmental conditions by building a 6-pollutant Delhi AQI dataset. Fourth, we implemented DLinear (AAAI 2023) as a rigorous baseline. And fifth, we successfully demonstrated **Flexible Multi-Horizon Forecasting**: reusing a SINGLE trained D2Vformer checkpoint across 48h, 72h, 96h, 192h, and 336h without any retraining!"

"Our empirical results revealed a key scientific finding: On Delhi AQI, where calendar cycles are dominant, D2Vformer consistently beats retrained DLinear across all horizons — achieving a **50.1% lower MSE at 336 hours** while requiring zero retraining time. On ETTh1, where linear physical

trends dominate, DLinear achieves lower error. Finally, we integrated our models into an offline Streamlit web dashboard for real-time visualization. Thank you, sir. We are ready for your questions."

SECTION 33 — Comprehensive Viva Preparation (42 Questions)

Har question ke teen levels hain: **Easy Hinglish**, **Technical English**, aur **One-Line Viva Punchline**.

Q1: What is your project about?

Easy: Hamara project D2Vformer architecture use karke time series data (jaise Delhi pollution aur Transformer temperature) ka future predict karta hai, jisme single model bina retraining ke multiple future horizons predict kar sakta hai.

Technical: We implemented and evaluated D2Vformer, a time-series forecasting model that leverages Date2Vec periodic calendar embeddings and cross-attention Fusion Blocks to enable flexible multi-horizon forecasting without retraining.

One-Liner: *"Flexible multi-horizon time-series forecasting using calendar-aware Transformer embeddings."*

Q2: What is the novelty of D2Vformer?

Easy: Purane models time steps ko 1, 2, 3 ki tarah dekhte hain aur ek fixed horizon ke liye train hote hain. D2Vformer real calendar dates ko Fourier vectors bana kar future timestamp ko query banata hai, jisse ek hi checkpoint multiple horizons predict kar sakta hai.

Technical: Decoupling the forecasting horizon from the historical sequence length via Fourier-based Date2Vec timestamp embeddings and Query-based cross-attention in the Fusion Block.

One-Liner: *"Date2Vec calendar embeddings enabling zero-shot variable horizon forecasting."*

Q3: Why did you choose DLinear as your baseline?

Easy: DLinear AAAI 2023 ka famous paper tha jisne prove kiya tha ki complex transformers aksar simple linear models se haar jaate hain. Humne dekhna chaha ki kya D2Vformer sach mein linear model se accha hai.

Technical: DLinear is the recognized competitive benchmark for long-sequence time series (AAAI 2023), decomposing series into trend and seasonal components via moving averages and linear projections.

One-Liner: *"To provide an honest, rigorous sanity-check against standard linear trend-seasonal decomposition."*

Q4: Why does DLinear beat D2Vformer on ETTh1, but loses on Delhi AQI?

Easy: Transformer oil temperature physical inertia aur linear continuous trend par chalta hai jahan calendar ka koi lena-dena nahi hota. Lekin Delhi pollution mein rush hours, weekends, aur winter seasons jaise calendar events dominant hote hain jahan Date2Vec jeet jata hai.

Technical: ETTh1 exhibits strong autocorrelation and autoregressive trends favored by linear projections, whereas Delhi AQI has strong diurnal and seasonal calendar periodicity perfectly captured by Date2Vec's Fourier representations.

One-Liner: *"Domain characteristics: ETTh1 is autoregressive/trend-dominated, whereas Delhi AQI is calendar-periodicity-dominated."*

Q5: What does "without retraining" mean?

Easy: Iska matlab hai ki humne model ko sirf 96h par ek baar train kiya. Phir 48h, 72h, 192h, aur 336h predict karne ke liye humne weights ko 0.001% bhi update nahi kiya — wahi saved checkpoint reuse kiya.

Technical: Model parameters remain strictly frozen in evaluation mode; the architecture dynamically ingests variable numbers of future timestamp queries to output variable-length horizons.

One-Liner: *"Inference across variable horizon lengths using a single frozen parameter checkpoint."*

Q6: What were the 6 bugs you fixed in the released repository?

Easy: Pandas 3.0 apply error, NumPy 2.0 np.Inf deletion, missing self.verbose, missing --save_path, missing --mark_index, aur Fusion Block mein dimension mismatch ($x + y$ ki jagah $V + y$).

Technical: Handled modern library deprecations in Pandas/NumPy, corrected uninitialized experiment attributes, added missing CLI parser arguments, and resolved tensor broadcasting shape mismatch in the Fusion Block residual addition.

One-Liner: *"Three environment deprecation fixes, two CLI argument patches, and one structural tensor dimension mismatch correction."*

Q7: What is seq_len and pred_len?

Easy: seq_len = 96 ghante ka pichla history data jo model dekhta hai. pred_len = aage ke ghante jo model predict karta hai.

Technical: seq_len is the historical input lookback window length; pred_len is the output target forecast horizon.

One-Liner: *"seq_len is historical lookback; pred_len is future forecast length."*

Q8: What is Date2Vec?

Easy: Ek mathematical layer jo timestamp (hour, day, weekday, month) ko sine aur cosine waves ke combination mein convert karke vector bana deti hai.

Technical: A Fourier-based continuous time embedding module that decomposes discrete calendar features into multi-frequency trigonometric representations.

One-Liner: *"Fourier sine/cosine calendar timestamp feature embedder."*

Q9: Why use RevIN (Reversible Instance Normalization)?

Easy: Time series mein data ka mean aur variance waqt ke sath badal jaata hai (distribution shift). RevIN input par local normalization karta hai aur output par use wapas denormalize karta hai.

Technical: RevIN mitigates temporal distribution shift by symmetrically normalizing input sequences and restoring learned affine distributions at the model output.

One-Liner: *"Prevents temporal distribution shift between train and test windows."*

Q10: Why did you use D2Vformer_simple instead of the full model?

Easy: Full released model ke andar repo ke authors ke do aur unfixable bugs the (4D tensor patch mismatch aur linear projection layer ka call na hona). Simple variant ne core Date2Vec aur Fusion block ko bina bugs ke perfectly run kiya.

Technical: The full released implementation suffered from unresolvable architectural inconsistencies between 4D patched tensors and 3D Fusion blocks, whereas D2Vformer_simple correctly realized the core attention-based date-matching formulation.

One-Liner: *"To avoid unresolvable tensor broadcasting defects present in the author's uncured full model release."*

(Questions 11 to 42 cover early stopping, patch length, loss formulation, Colab deployment, Streamlit architecture, RMSE vs MAE, etc., as documented throughout the preceding chapters).

SECTION 34 — Common Confusions Solved

Things You Are Likely to Confuse During Stress:

- **seq_len vs pred_len:** seq_len input history hai (hamesha 96), pred_len output forecast hai (48, 72, 96, 192, 336).
- **Normalized MSE vs Denormalized MSE:** Research table ka MSE (0.1179) normalized space mein hai; Streamlit ka live window MSE (332.25) physical units squared mein hai. Dono bilkul accurate hain.
- **D2Vformer Checkpoints vs DLinear Checkpoints:** D2Vformer ka 1 checkpoint har horizon ke liye reuse hota hai. DLinear ke 5 alag-alag checkpoints hain kyunki uski projection layer fixed size ki hoti hai.
- **Delhi AQI Data Source:** Yeh physics-informed synthetic dataset hai jo CPCB distribution aur seasonal patterns par calibrated hai, live sensor pipeline nahi.
- **Semester 1 vs Semester 2:** Semester 1 = reproduction, 6 bug fixes, sweep, Delhi AQI, DLinear, flexible 5-horizon demo. Semester 2 = arbitrary sparse queries, uncertainty estimation, PatchTST.

SECTION 35 — One-Page Revision Cheat Sheet

5-Minute Pre-Review Cram Sheet

Project Title	D2Vformer Time-Series Forecasting (IEEE TNNLS 2026)
Core Innovation	Date2Vec (Fourier calendar embedding) + Fusion Block (cross-attention)
Primary Claim	Single trained checkpoint reusability across multiple horizons without retraining
Datasets	ETTh1 (7 features, Oil Temp target) Delhi AQI (6 pollutants: PM2.5, PM10, NO2, SO2, CO, O3)
Key Parameters	seq_len=96, label_len=48, lr=0.001, batch_size=64, T2V_outmodel=36, patience=5
Tested Horizons	48h (2 days), 72h (3 days), 96h (4 days), 192h (8 days), 336h (14 days)
Baseline Model	DLinear (AAAI 2023) — Moving average trend + seasonal linear projections
ETTh1 Outcome	DLinear achieves lower error across all horizons (trend-dominated dynamics)
Delhi AQI Outcome	D2Vformer beats DLinear across all horizons (13.8% to 50.1% MSE advantage at 336h)
Retraining Cost	D2Vformer: 0 seconds (reused) DLinear: ~299 seconds (10 separate checkpoints)
Bugs Fixed	6 bugs across 7 files (Pandas 3.0, NumPy 2.0, verbose, save_path, mark_index, Fusion residual)
Demo Platform	Streamlit interactive web dashboard running 100% offline locally

SECTION 36 — A-to-Z Technical Glossary

Term	Simple Hinglish Meaning	Project Meaning
Attention	Dhyan dena! Kaunsa past data future query se match karta hai.	Dot-product similarity weighting between Query (future date) and Key (history date).
Baseline	Standard comparison benchmark model.	DLinear (AAAI 2023) individual channel decomposition model.
Checkpoint	Trained model ki saved file (<code>.pk1</code>).	PyTorch state_dict saving best validation weights.
Date2Vec	Calendar timestamp ko vector banane wala module.	Fourier sine/cosine continuous temporal representation layer.
Early Stopping	Overfitting se pehle training rok dena.	Patience=5 epochs monitor on validation loss.
Fusion Block	Future query aur past data ko attention se jodne wala block.	Cross-attention block outputting future forecast sequences.
Horizon	Kitna aage ka future predict karna hai.	pred_len (48h, 72h, 96h, 192h, 336h).
Inference	Trained model se prediction nikalna.	Read-only evaluation forward pass without gradients.
Patching	Continuous sequence ke tukde (windows) banana.	Unfolding 96h sequence into 16-hour overlapping patches (stride=8).
RevIN	Reversible Instance Normalization.	Local mean/variance normalization removing distribution shift.

THE 20 ABSOLUTE MUST-KNOWS BEFORE YOUR REVIEW

In 20 baton ko review room mein ghusne se pehle 100% memorize kar lo:

1. **Project Purpose:** Flexible multi-horizon time series forecasting using calendar-aware Transformer embeddings.
2. **The Core Problem:** Traditional forecasting models require complete retraining whenever the requested forecast horizon length changes.
3. **What D2Vformer Is:** An IEEE TNNLS 2026 paper combining Fourier Date2Vec embeddings with attention-based Fusion Blocks.

4. **What Date2Vec Does:** Converts raw calendar timestamps (hour, day, weekday, month) into continuous sine/cosine periodic frequency vectors.
5. **What the Fusion Block Does:** Uses future timestamp embeddings as Queries to cross-attend over historical attention patterns.
6. **The Flexible Breakthrough:** A single D2Vformer checkpoint trained at 96h is reused across 48h, 72h, 96h, 192h, and 336h without retraining.
7. **Why DLinear Needs 10 Checkpoints:** DLinear's output projection matrix `Linear(96, H)` has fixed dimensions tied to horizon H , requiring separate retraining for each horizon.
8. **Delhi AQI Finding:** D2Vformer beats DLinear across all horizons on Delhi AQI (up to 50.1% MSE reduction at 336h) because air pollution is heavily governed by cyclic calendar patterns.
9. **ETTh1 Finding:** DLinear beats D2Vformer on ETTh1 because transformer oil temperature is dominated by linear physical autocorrelation and continuous trends.
10. **The Flexibility vs Accuracy Trade-off:** DLinear offers high point accuracy on trend datasets but requires N training pipelines. D2Vformer offers zero-retraining operational reusability and superior performance on calendar-driven environmental data.
11. **Our Reproduction Engineering:** We debugged and fixed 6 critical bugs in the official code across NumPy 2.0, Pandas 3.0, and Fusion Block residual tensor broadcasting.
12. **Why D2Vformer_simple:** The full released repository contained unresolvable 4D-vs-3D tensor mismatches and missing projection calls; the simple variant implements the true core innovation without defects.
13. **Paper Benchmark Discrepancy:** The paper's 0.44 MSE on ETTh1 was reported for the unreleased full model; our verified 0.6982 MSE corresponds to the stable simple variant.
14. **Delhi AQI Dataset Origin:** A physics-informed synthetic dataset calibrated to real CPCB Delhi pollution distributions across 6 pollutants (PM2.5, PM10, NO2, SO2, CO, O3).
15. **Our Hyperparameter Sweep:** 10 configurations on Google Colab T4 GPU identified $LR=0.001$ and $T2V=36$ as the optimal setup.
16. **Normalized vs Denormalized Metrics:** Test set tables use normalized Z-score metrics (0.1179); Streamlit displays denormalized physical units ($\mu\text{g}/\text{m}^3$ squared: 332.25) for single windows.
17. **Our Streamlit Demo:** An offline, locally running interactive UI displaying real calendar dates, seasons, horizon sliders, and live dual-model forecasts.
18. **Semester 1 Scope:** Code repair, sweep, India AQI, DLinear baseline, flexible evaluation, single-checkpoint reuse, Streamlit demo.
19. **Semester 2 Boundaries:** Arbitrary sparse irregular queries, uncertainty quantification, and PatchTST are future work and NOT yet completed.
20. **How to Answer Guide Critiques:** DLinear beating D2Vformer is not a failure — it validates established literature (AAAI 2023) and highlights the exact operational trade-off of zero-retraining horizon flexibility.